

	Work Instruction	Doc. Revision	00
	AXI V810 XXL Valve Assembly Removal & Replacement Procedure	Effective Date	30 May 2017

This procedure is applicable to below models:

V810 STD	☐	S1	☐	S2
V810 XXL	☒	S1	☒	S2
V810 S2EX	☐			
V810 Mini	☐			

REVISION HISTORY

Rev	Effective Date	Description	Doc Originator
00	30 May 2017	First Release	Chin Chee Yan

	Work Instruction	Doc. Revision	00
	AXI V810 XXL Valve Assembly Removal & Replacement Procedure	Effective Date	30 May 2017

Content

Removal Procedure.....	3
Replacement Procedure.....	5
Validation Procedure.....	11

	Work Instruction	Doc. Revision	00
	AXI V810 XXL Valve Assembly Removal & Replacement Procedure	Effective Date	30 May 2017

Removal Procedure

1. **Shutdown and completely power off machine.** Turn off air pressure regulator
2. **Power off the SSCA main power # Important**
3. Open front access door and manual move XY stage towards front door direction

Remarks:

- Equipment SN MY13138010 & below equip with Inch Size Inlet Port at Valve Assembly (Figure 1)
- Equipment SN MY13138011 & above equip with Metric Size Inlet Port at Valve Assembly (Figure 1)

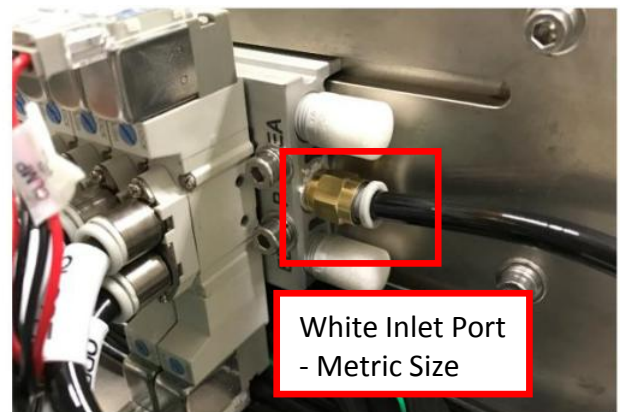
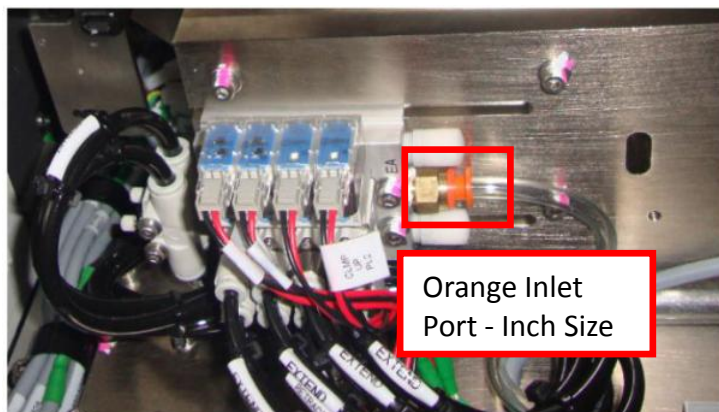


Figure 1 Inch and Metric Inlet Port

4. Press inlet port and remove the main tubing as Figure 2



Figure 2 XXL Valve Assembly

	Work Instruction	Doc. Revision	00
	AXI V810 XXL Valve Assembly Removal & Replacement Procedure	Effective Date	30 May 2017

5. Disconnect 6 x Solenoid Valve Cables as Figure 3 & 4 .Remove 8 x tubing pull as below Figure 3.
Remove necessary cable tie as well. Remarks : Perform labeling at 8 x Tubing upon removing

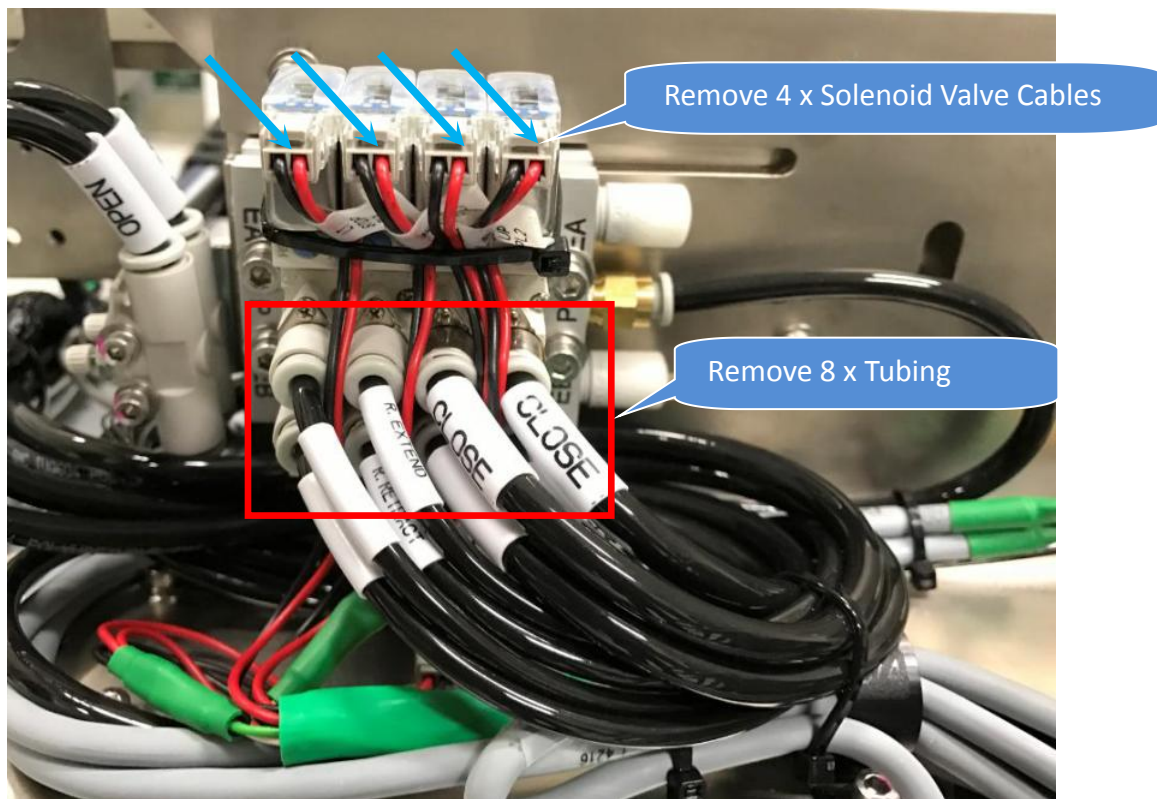


Figure 3 Cable & Tubing Removal

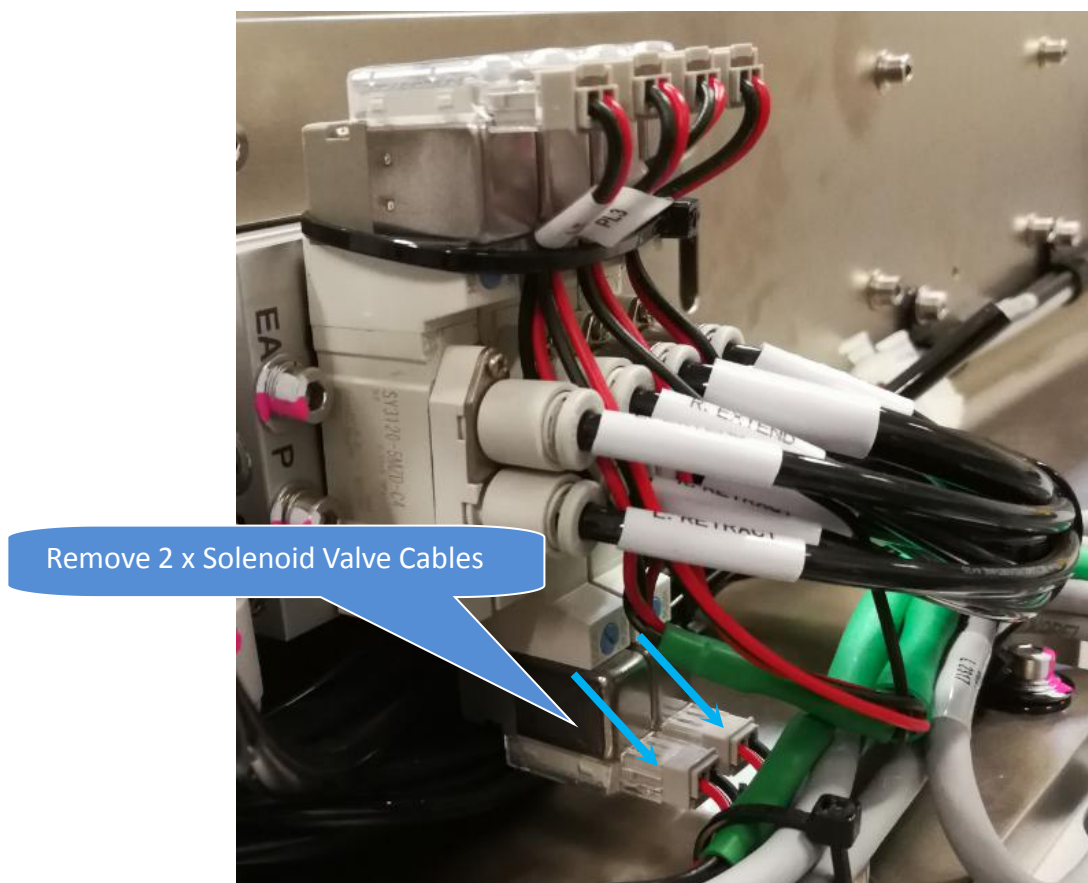


Figure 4 Cable Removal

ViTrox	Work Instruction	Doc. Revision	00
	AXI V810 XXL Valve Assembly Removal & Replacement Procedure	Effective Date	30 May 2017

6. Remove 4 x Screws and remove entire XXL Valve Assembly from Fix Rail. Remove necessary cable tie

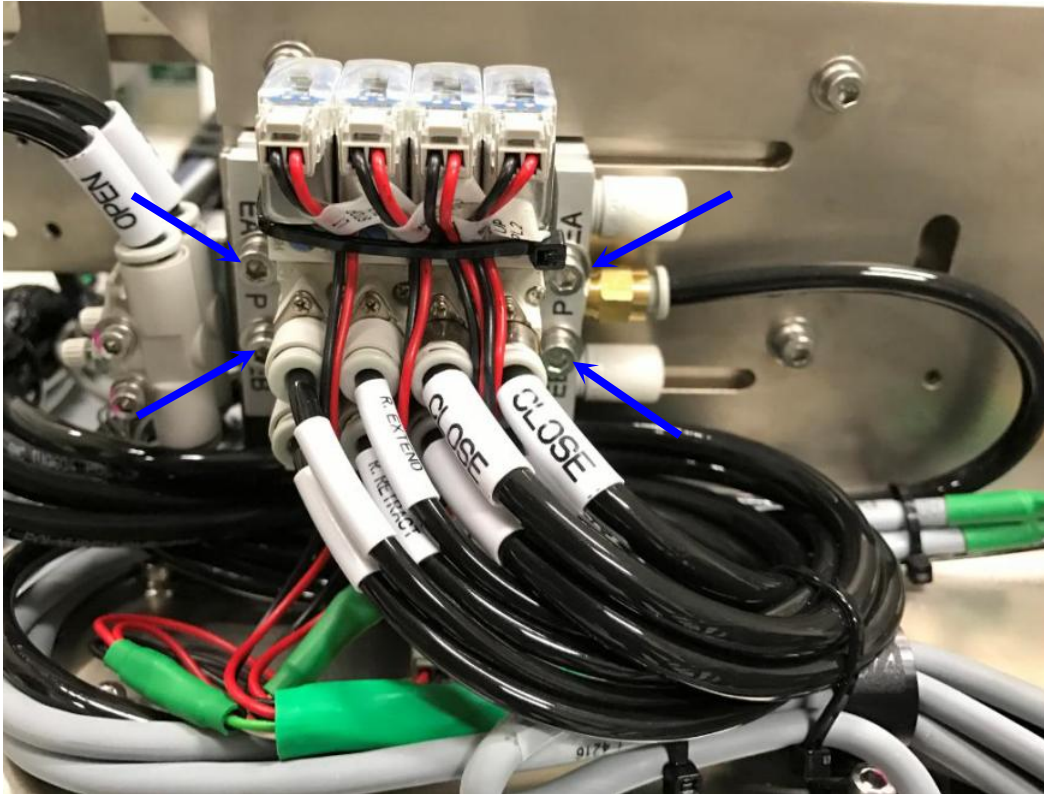


Figure 5 XXL Valve Assembly Removal

Replacement Procedure

7. Prepare new XXL Valve, Assembly (P/N, 4XASP-A087) as shown.

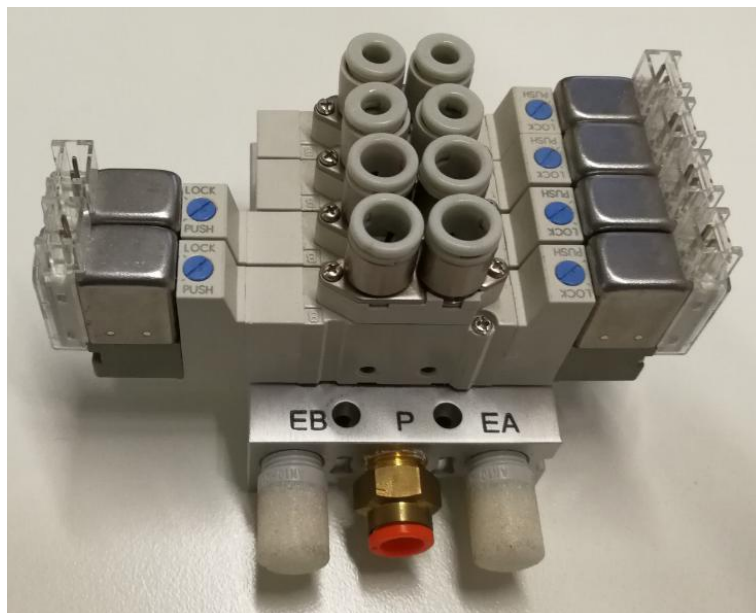


Figure 6 XXL Valve Assembly

ViTrox	Work Instruction	Doc. Revision	00
	AXI V810 XXL Valve Assembly Removal & Replacement Procedure	Effective Date	30 May 2017

a. Equipment SN MY13138010 & below (Inch Size Inlet Port at Valve Assembly)

i. Direct refer to Step 8

b. Equipment SN MY13138011 & above equip with Metric Size Inlet Port at Valve Assembly
(Figure 1)

i. Remove Inch Inlet Port from new XXL Valve, Assembly (P/N, 4XASP-A087)

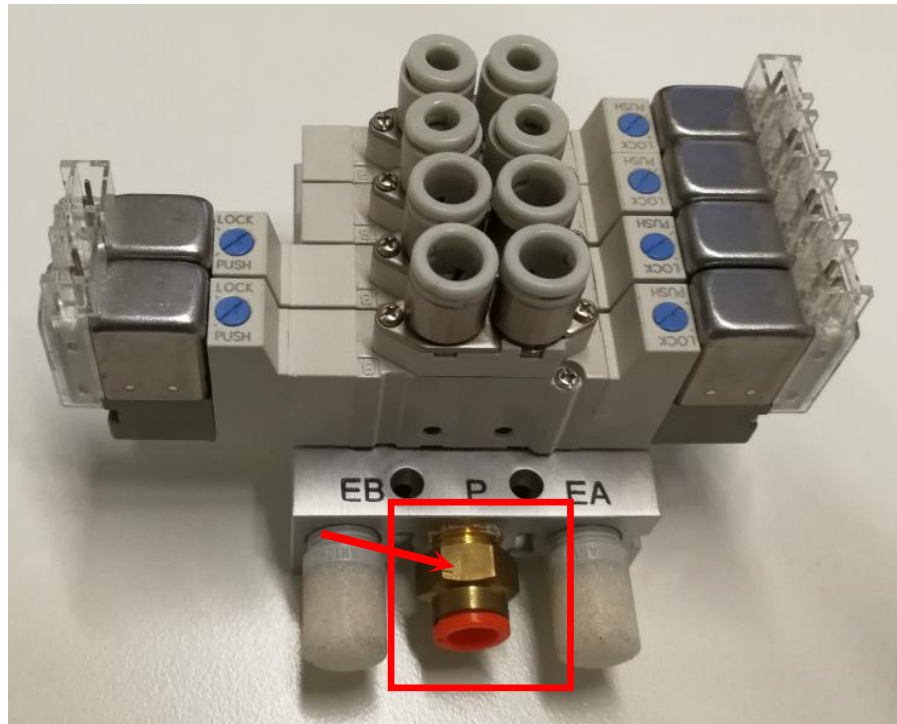


Figure 7 XXL Valve Assembly

ii. Remove Existing Metric Inlet Port from XXL Valve Assembly (Previously dismantle)

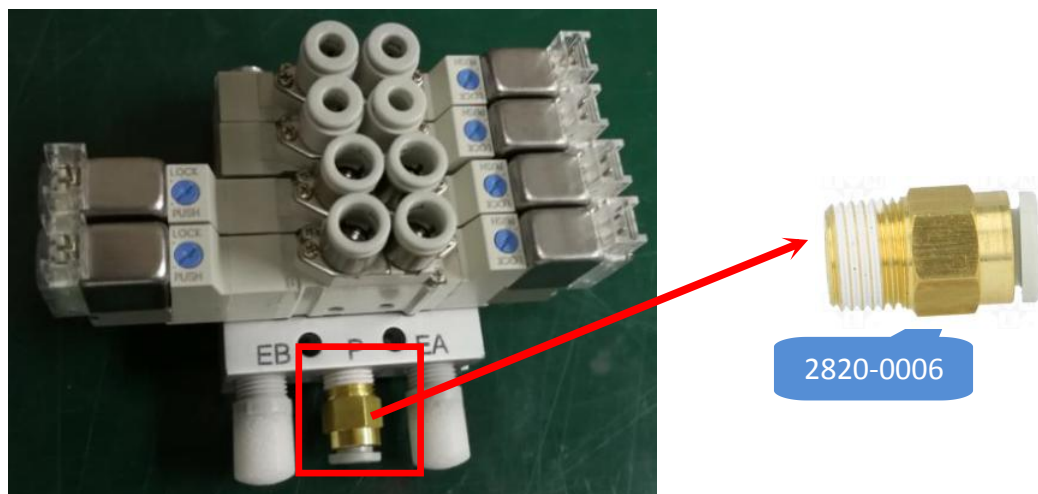


Figure 8

	Work Instruction	Doc. Revision	00
	AXI V810 XXL Valve Assembly Removal & Replacement Procedure	Effective Date	30 May 2017

- iii. Assemble the Existing Metric Inlet Port at New Valve, Assembly (P/N, 4XASP-A087) as below Figure 8

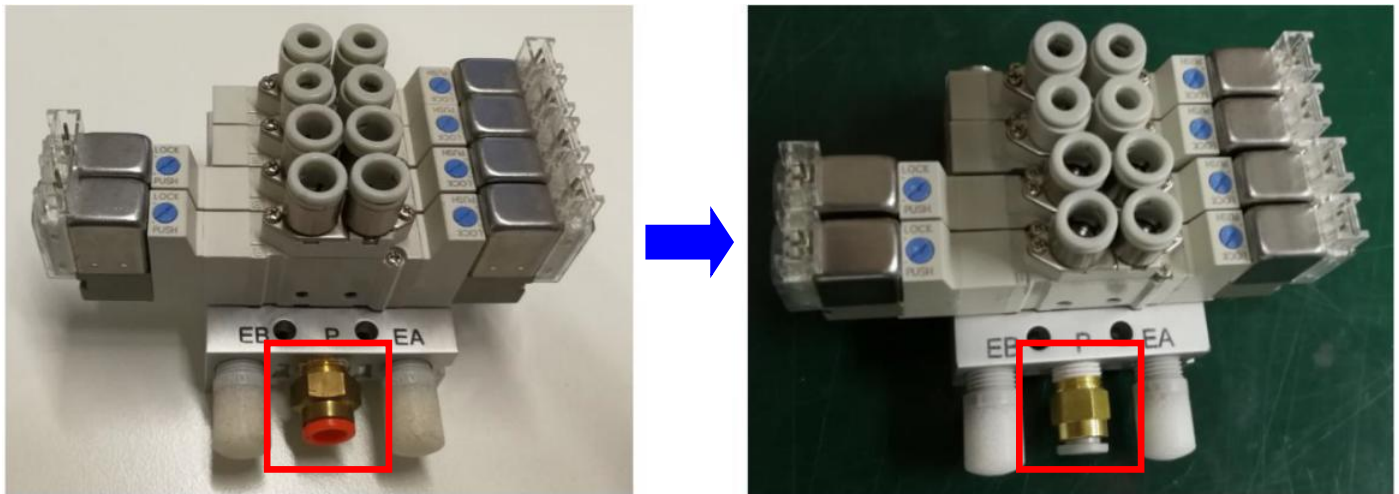


Figure 9 Metric Inlet Port Swapping

8. Replace new 4XASP-A087 (XXL Valve Assembly) at Fix Rail. Tighten 4 x Screws as below Figure 9

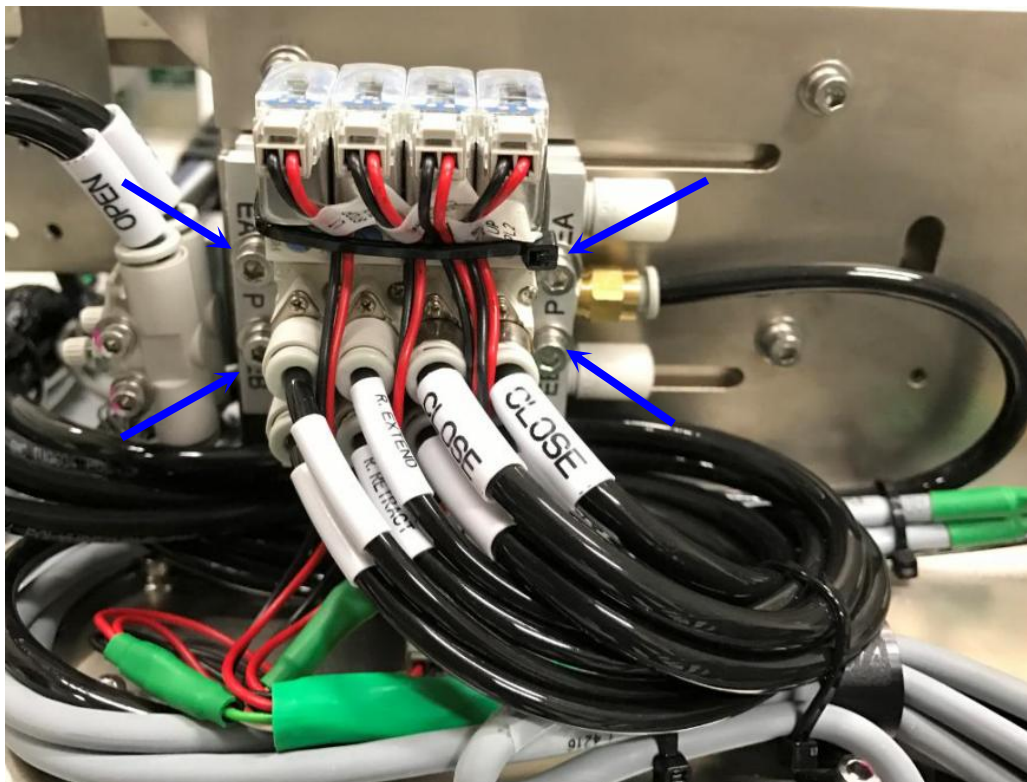


Figure 10 XXL Valve Assembly

9. Connect 6 x Solenoid Valve Cables (Figure 11 &12) and 8 x Tubing to origin position by refer to Table 1.
1. Apply cable tie as shown.

Pneumatic Valve				
Solenoid Valve Cables	LT Stop (PL3)	RT Stop (PL1)	CLP Up (PL2)	CLP Up (PL2)
Tubing	Extend	Extend	Close	Close
Tubing	Retract	Retract	Open	Open
Solenoid Valve Cables	NA	NA	CLP Down (PL9)	CLP Down (PL9)

Table 1

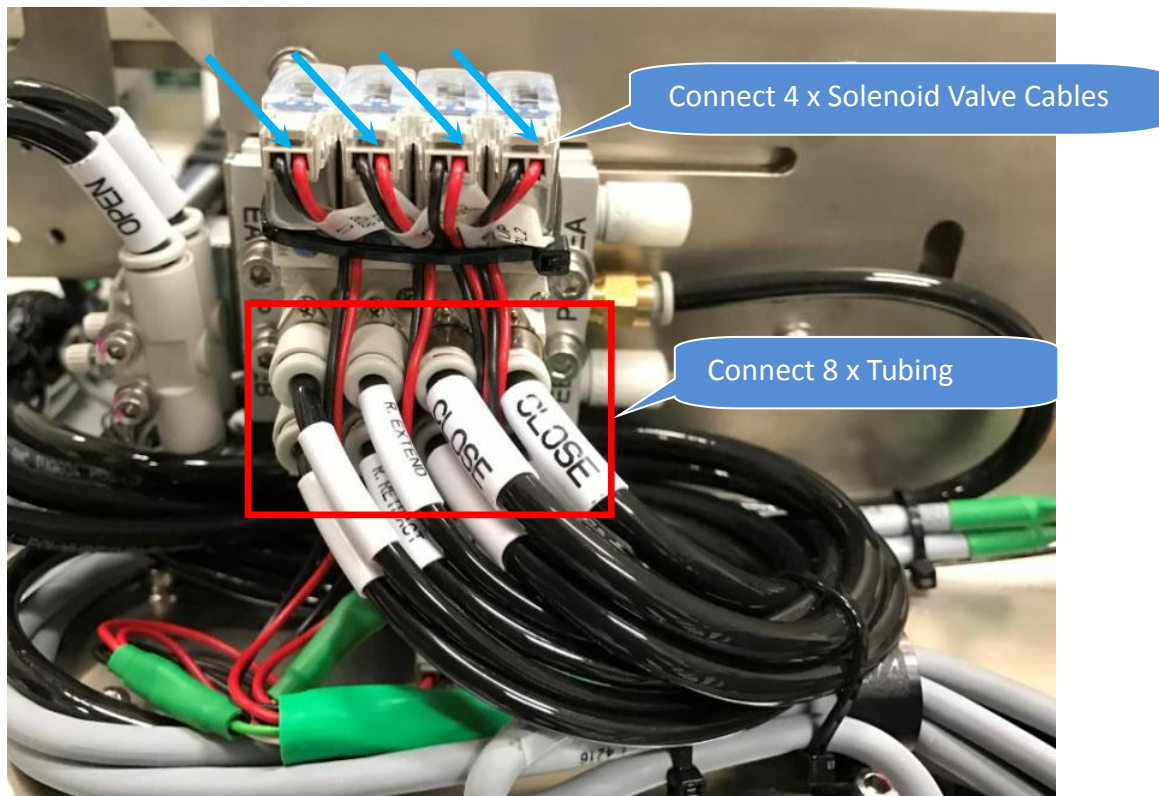


Figure 11 Cable & Tubing Removal

	Work Instruction	Doc. Revision	00
	AXI V810 XXL Valve Assembly Removal & Replacement Procedure	Effective Date	30 May 2017

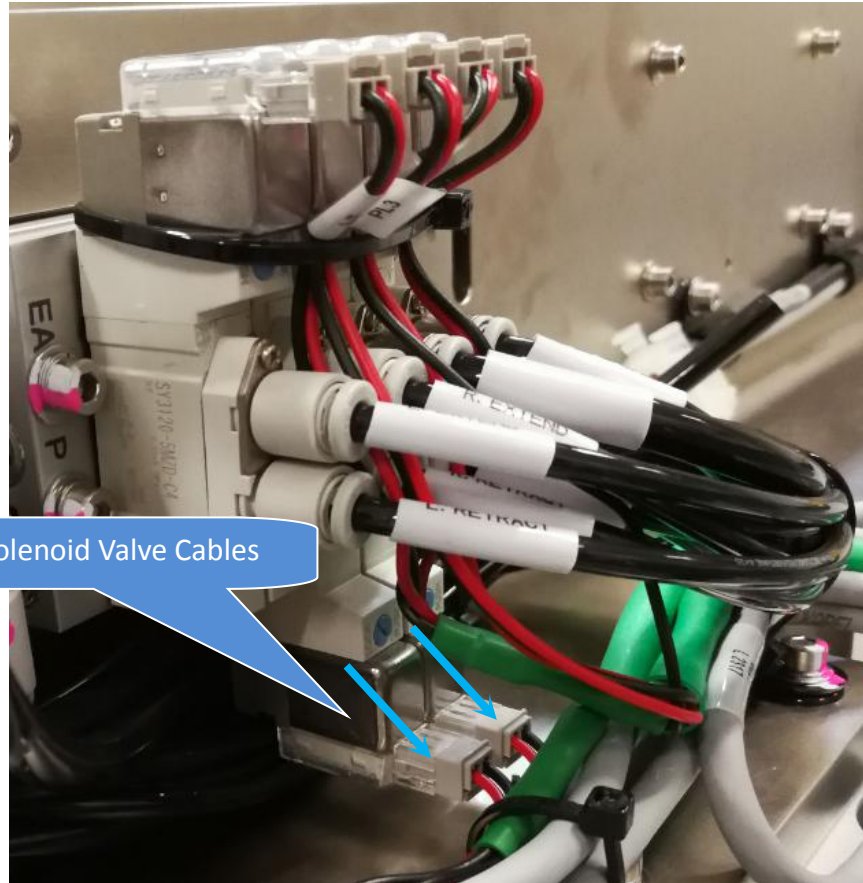


Figure 12 Cable Removal

10. Plug the main input tubing into main inlet port as below Figure 13.

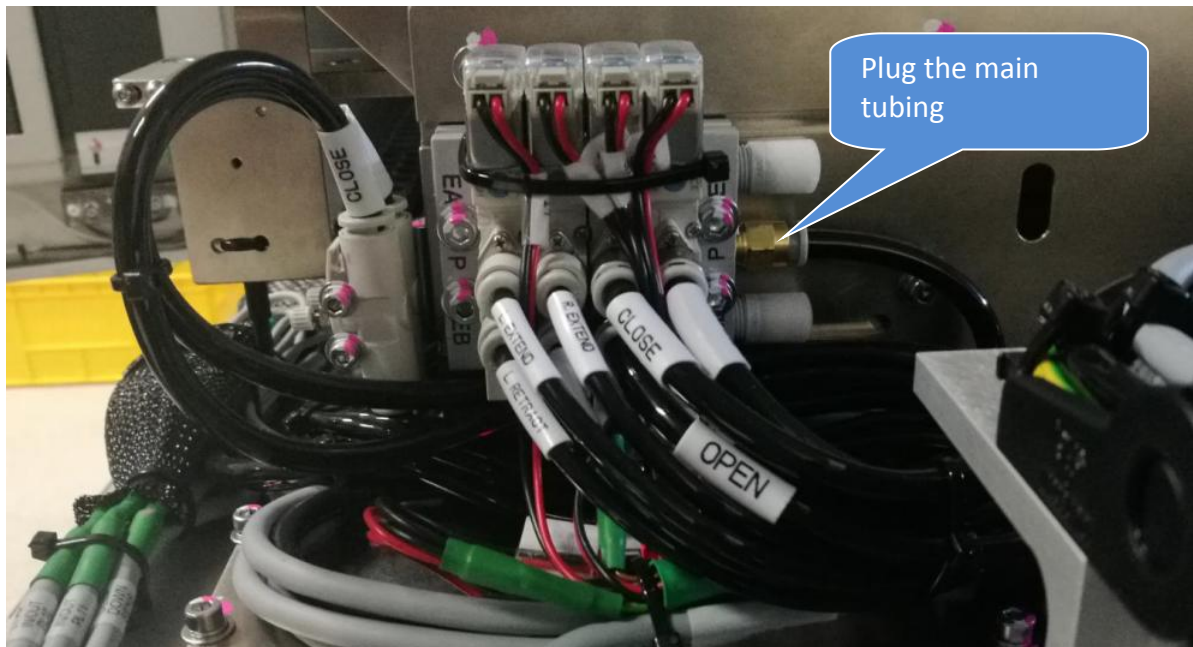
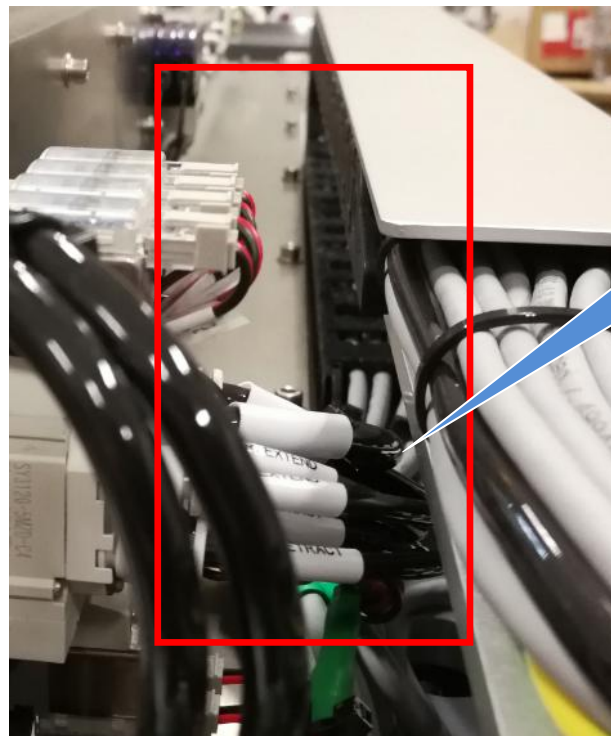


Figure 13 Main tubing Connect

ViTrox	Work Instruction	Doc. Revision	00
	AXI V810 XXL Valve Assembly Removal & Replacement Procedure	Effective Date	30 May 2017

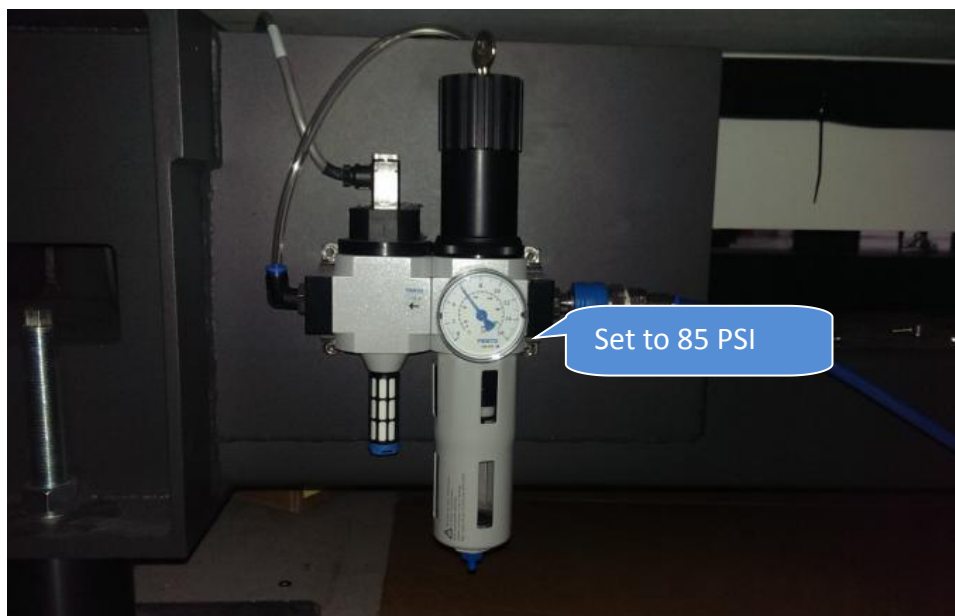
11. Gently move X axis stage and visual check where Tubing & Solenoid cable do not contain friction with E-Chain Guard Plate.



No friction
between Valve
Assembly & X
Axis E-Chain

Figure 14 XXL Valve Assembly (Visual Check)

12. Remove all tools from machine, turn on main air regulator and start-up machine as usual.



Set to 85 PSI

Figure 15 Air Pressure Regulator adjustment

ViTrox	Work Instruction	Doc. Revision	00
	AXI V810 XXL Valve Assembly Removal & Replacement Procedure	Effective Date	30 May 2017

Validation Procedure

13. Log in V810 GUI, go to panel handling and click initialize. Turn off X-ray and open machine front access door. Next, go to Digital IO Tab By refer to Step E,F, G and H, visual check on mechanism movement, ensure actual movement is sync with the command toggle.

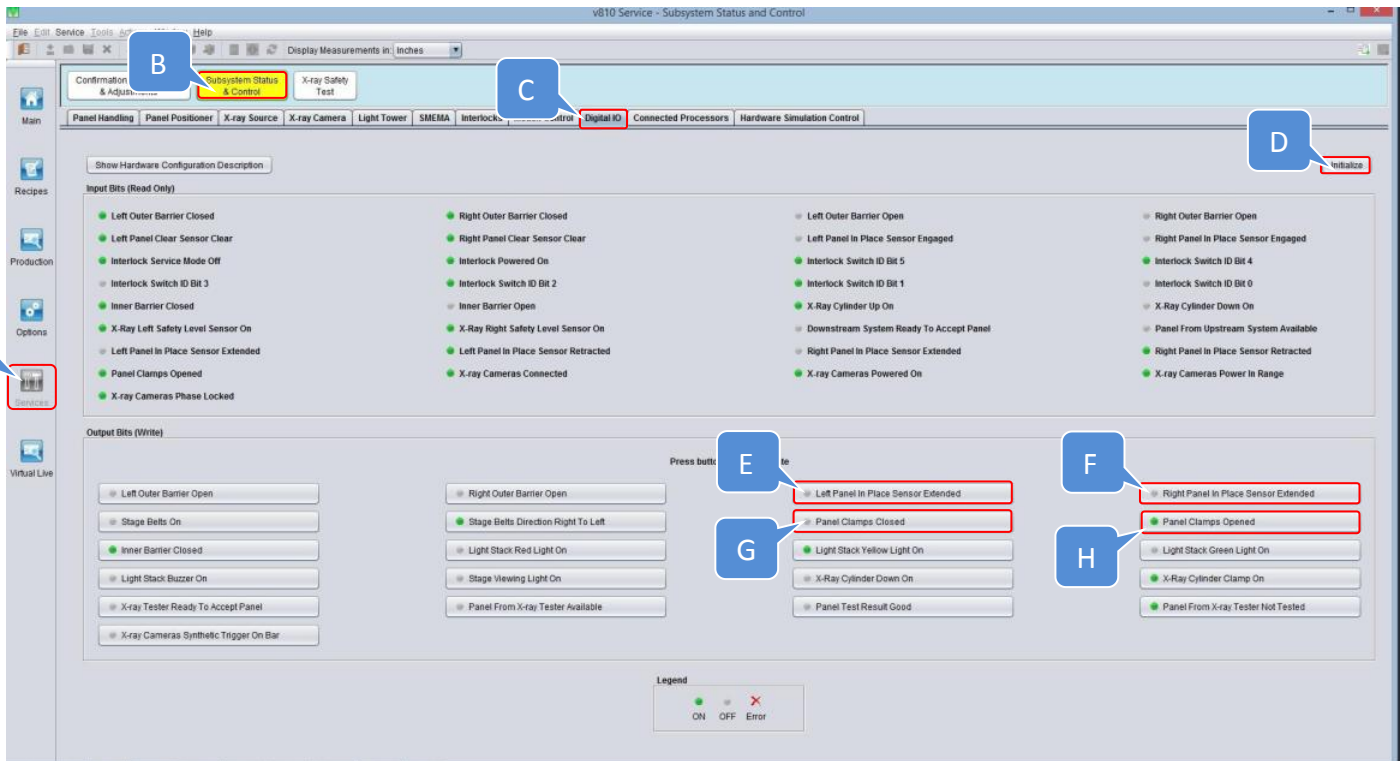


Figure 16 Panel Handling Mechanism Validation (Digital IO)

14. Close front access door and start-up as usual.